Does a Wavelet Front-End Help a ResNeXt Segment Roads?

A Controlled Ablation on Massachusetts Roads and DeepGlobe

Technical report: architecture, equations, protocol and results

October 2026

Abstract

We test, under controlled conditions, whether decomposing the input of a ResNeXt-50 road segmentation network with a single-level Haar discrete wavelet transform (DWT) improves road extraction. Four model variants ("rungs") share one architecture, one training recipe and one set of seeds, and differ only in how spatial resolution is reduced and restored: R0, a stock ImageNet-pretrained ResNeXt-50 U-Net; R1, the same network fed a 12-channel Haar decomposition (the proposed method); R2, learned Haar-basis pooling and unpooling at every resolution change; and R3, R2 trained with an added soft-clDice topology loss. U-Net and D-LinkNet serve as external baselines. We give an exact, function-preserving initialisation of the DWT-input stem from the pretrained weights, which makes R1 identical to R0 at the start of training, and show that a DWT followed by a convolution is a reparameterisation of a strided convolution. Across 36 runs (6 models $\times$ 2 datasets $\times$ 3 seeds) and a pre-specified analysis with a seed-aware bootstrap, the primary comparison R1 $-$ R0 in per-image test IoU is $+0.0004$ (95% CI $[-0.0040, +0.0060]$) on Massachusetts Roads and $+0.0017$ ($[-0.0018, +0.0059]$) on DeepGlobe: no detected difference on either dataset.

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1 Question and design

The proposal under test decomposes each colour channel of the input with a single-level 2D Haar DWT, stacks the resulting 12 sub-bands, and feeds them to a ResNeXt encoder whose first convolution is widened from 3 to 12 input channels. The claim is that the sub-bands expose edge and texture information that helps the network find thin, occluded roads.

To make the effect of the DWT measurable, every variant below shares the encoder backbone, the decoder, the training data stream, the optimiser, the schedule, the loss (except R3) and the evaluation. They differ only in the operators that change spatial resolution.

Table 1: The ablation ladder. All rungs use an ImageNet-pretrained ResNeXt-50 32×4d encoder and the same U-Net decoder.

Rung	Encoder downsampling	Decoder upsampling	Loss	Role
R0	strided conv + max-pool (stock)	bilinear + 3×3 conv	BCE + Dice	control
R1	Haar DWT at the input, stride-1 stem	bilinear + 3×3 conv	BCE + Dice	**proposed method**
R2	learned Haar-basis pooling everywhere	learned Haar-basis unpooling + 3×3 conv	BCE + Dice	strong wavelet variant
R3	as R2	as R2	BCE + Dice + $\frac{1}{2}$clDice	connectivity variant
U-Net	—	—	BCE + Dice	external baseline (from scratch)
D-LinkNet	—	—	BCE + Dice	external baseline (ResNet-34)

Pre-specified primary analysis. The primary quantity is R1 − R0 in per-image test IoU at the validation-selected threshold, estimated with a two-level bootstrap that resamples seeds within each arm and then images (Section 6). A difference is claimed only if both this interval and a seed-paired t interval exclude zero with the same sign. Everything else is secondary and reported without multiplicity correction. The plan and the rule were committed to the code repository before any test score was read.

2 Datasets

2.1 Massachusetts Roads

Aerial RGB imagery of Massachusetts at 1 m per pixel, in 1500 × 1500 tiles with binary road masks rasterised from OpenStreetMap centre-lines [1]. The official split is used: 1108 training, 14 validation and 49 test tiles.

- Original source (used here): `https://www.cs.toronto.edu/~vmnih/data/`
- Kaggle mirror: `https://www.kaggle.com/datasets/balraj98/massachusetts-roads-dataset`

No-data regions. Many tiles contain large blank white areas where no imagery exists. In a 40-image sample they covered 11% of pixels on average (over 30% in 7 images, up to 79%), and the labels have roads drawn through them. A pixel is treated as no-data if all three channels are ≥ 250 *and* it belongs to a 4-connected white region of at least 2500 pixels; isolated bright pixels (roofs, lane paint) are kept. The resulting validity mask V is applied to the training loss, to crop sampling, to checkpoint and threshold selection, and to every metric. Consequently Massachusetts scores here are *not* directly comparable with published numbers that count those pixels.

2.2 DeepGlobe Road Extraction

Satellite RGB imagery at 0.5 m per pixel from Thailand, Indonesia and India, in 1024×1024 tiles [3]. Only the 6226 training tiles of the challenge were released with masks; the challenge's validation and test tiles have no public labels. We therefore re-split the 6226 labelled tiles by image into 70/15/15% with a fixed seed (1337): 4358 training, 933 validation and 935 test tiles. The test split is pinned by the SHA-256 of its sorted, newline-joined tile ids:

`a18e6da5754390b186234dbe6ba98357 a1618c9a03d12688a0bdc4fbf65883fe`

DeepGlobe has no no-data regions, so $V \equiv 1$.

- Challenge page: `http://deepglobe.org/`
- Kaggle release: `https://www.kaggle.com/datasets/balraj98/deepglobe-road-extraction-dataset`
- Mirror used here (no Kaggle credentials were available): `https://huggingface.co/datasets/lammtfkday/deepglobe-roads`. Its counts, tile size and resolution match the original; pixel identity with the Kaggle release was not verified. The mirror's own 5000/600/626 partition is not used.

Table 2: Data used in the experiments.

Dataset	Tile size	GSD	Train / val / test	Road pixels	No-data masking
Massachusetts Roads	1500^2	1.0 m	1108 / 14 / 49	≈5% of valid	yes
DeepGlobe	1024^2	0.5 m	4358 / 933 / 935	≈3.6%	no

No resizing. Imagery is never resized: roads are 2–4 pixels wide, and downsampling would delete exactly the structures under study. Training uses 512×512 crops; evaluation tiles each image and stitches the predictions back to full resolution (Section 5).

3 Architecture

3.1 The Haar transform as fixed convolutions

Let $\ell = \frac{1}{\sqrt{2}}(1, 1)$ and $h = \frac{1}{\sqrt{2}}(1, -1)$. The four 2×2 analysis kernels are the outer products

$$\kappa_{LL} = \ell^\top \ell, \quad \kappa_{LH} = h^\top \ell, \quad \kappa_{HL} = \ell^\top h, \quad \kappa_{HH} = h^\top h, \tag{1}$$

so for a 2×2 pixel block $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ the coefficients are

$$\mathrm{LL} = \tfrac{a+b+c+d}{2}, \quad \mathrm{LH} = \tfrac{a+b-c-d}{2}, \quad \mathrm{HL} = \tfrac{a-b+c-d}{2}, \quad \mathrm{HH} = \tfrac{a-b-c+d}{2}. \tag{2}$$

The four kernels form an orthonormal basis of $\mathbb{R}^{2\times 2}$. Applied channel-wise with stride 2,

$$\mathrm{DWT}: \ \mathbb{R}^{C\times H\times W} \to \mathbb{R}^{4C\times \frac{H}{2}\times \frac{W}{2}}$$

is a grouped convolution with frozen weights; channel order is $[c\text{-LL}, c\text{-LH}, c\text{-HL}, c\text{-HH}]$ for each input channel c. Its inverse IDWT is the adjoint (transposed) convolution with the same kernels and is exact: $\mathrm{IDWT}(\mathrm{DWT}(x)) = x$. Both run on the GPU inside the network.

3.2 Encoder

The encoder is ResNeXt-50 32×4d [4]. Each bottleneck block computes

$$y = x + \sum_{i=1}^{C} \mathcal{T}_i(x), \qquad C = 32, \tag{3}$$

implemented as a 1×1 reduction, a 3×3 convolution with 32 groups, and a 1×1 expansion. The cardinality $C = 32$ is the same in every stage; what grows per stage is the width. For a 512×512 input the five feature maps passed to the decoder are:

Feature	Produced by	Resolution	Channels
f_0	stem	256^2	64
f_1	layer1	128^2	256
f_2	layer2	64^2	512
f_3	layer3	32^2	1024
f_4	layer4 (bottleneck)	16^2	2048

3.3 R1: the DWT-input stem, and an exact initialisation

Stride parity. The DWT already halves the resolution, so R1's stem convolution has stride 1:

$$\begin{aligned}
&\text{R0:} \quad x \in \mathbb{R}^{3\times 512^2} \xrightarrow{\text{conv } 7\times 7,\ s=2} \mathbb{R}^{64\times 256^2} \xrightarrow{\text{pool},\ s=2} \mathbb{R}^{64\times 128^2} \\
&\text{R1:} \quad x \xrightarrow{\text{DWT}} \mathbb{R}^{12\times 256^2} \xrightarrow{\text{conv } 7\times 7,\ s=1} \mathbb{R}^{64\times 256^2} \xrightarrow{\text{pool},\ s=2} \mathbb{R}^{64\times 128^2}.
\end{aligned}$$

Every later feature map, skip connection and the whole decoder then have identical shapes in R0 and R1, and the output resolution equals the input resolution (asserted in the code).

Weight adaptation. Widening the stem from 3 to 12 input channels discards its ImageNet weights unless they are adapted. The natural adaptation — give each colour's LL band the pretrained kernel (halved, since LL is twice the local mean) and zero the other bands — is *not* equivalent to R0: it is the pretrained kernel applied at stride 1 to a 2×2-average-pooled image. An exact adaptation exists.

Proposition 1 (Exact DWT-domain stem)**.** *Let $W \in \mathbb{R}^{O\times 3\times 7\times 7}$ be R0's stem kernel (stride 2, padding 3). Let $K \in \mathbb{R}^{O\times 3\times 8\times 8}$ be W zero-padded by one row at the top and one column at the left, and $G = \mathrm{DWT}(K) \in \mathbb{R}^{O\times 12\times 4\times 4}$ (the same Haar analysis applied to the kernel). Define $W' \in \mathbb{R}^{O\times 12\times 7\times 7}$ by $W'[:, :, 1{:}5, 1{:}5] = G$ and zero elsewhere. Then for every input x with even height and width,*

$$\mathrm{conv}_{s=1,\,p=3}\big(\mathrm{DWT}(x);\, W'\big) \;=\; \mathrm{conv}_{s=2,\,p=3}(x;\, W). \tag{4}$$

Sketch. R0 computes, and after padding equivalently,

$$y_{o,i,j} = \sum_{c,u,v} W_{o,c,u,v}\, x_{c,\,2i+u-3,\,2j+v-3} = \sum_{c,p,q} K_{o,c,p,q}\, x_{c,\,2i+p-4,\,2j+q-4}, \qquad p, q \in \{0, \dots, 7\}.$$

Writing $p = 2m + r$, $q = 2n + s$ with $m, n \in \{0, \dots, 3\}$ and $r, s \in \{0, 1\}$, the sum runs over the 4×4 whole 2×2 blocks $\beta = (i + m - 2,\ j + n - 2)$ of the image. Within a block, the inner product of

the kernel block with the pixel block equals the inner product of their Haar coefficients, because the Haar basis is orthonormal. Hence

$$y_{o,i,j} = \sum_{c,k,m,n} G_{o,4c+k,m,n}\ \mathrm{DWT}(x)_{4c+k,\, i+m-2,\, j+n-2},$$

a 4×4 stride-1 convolution on the coefficient map; block offsets $-2, \ldots, +1$ are taps $1, \ldots, 4$ of a 7×7 kernel with padding 3. Zero padding agrees on both sides because, with even image dimensions, every block lies wholly inside or wholly outside the image. □

With this initialisation R1 is the *same function* as the pretrained R0 at step 0 (verified numerically: maximum logit difference 1.5×10^{-8} on the real launch path). R1 can depart from R0 only by learning; its stem has $12 \cdot 49 \cdot 64 = 37{,}632$ weights against R0's 9408, i.e. it can represent a 14×14-pixel footprint where R0 has 7×7.

Interpretation. Proposition 1 shows that "DWT, then convolution" can represent any strided convolution of the input, and conversely a stride-1 convolution over Haar sub-bands is a strided convolution with a larger footprint. The DWT front-end therefore does not add information; it changes the parameterisation and the receptive field of the first layer.

3.4 R2: learned Haar-basis pooling and unpooling

WaveletPool. For $x \in \mathbb{R}^{C \times H \times W}$ with sub-bands $D = \mathrm{DWT}(x) \in \mathbb{R}^{4C \times \frac{H}{2} \times \frac{W}{2}}$,

$$\mathrm{WP}(x)_c = \sum_{k=0}^{3} \omega_{c,k}\, D_{4c+k}, \qquad \omega \in \mathbb{R}^{C \times 4}, \tag{5}$$

a learned per-channel 2×2 analysis filter expressed in the Haar basis ($4C$ parameters, no channel mixing). By Eq. (2), the initialisation $\omega_c = (\frac{1}{2}, 0, 0, 0)$ gives 2×2 average pooling, and $\omega_c = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ gives $\frac{1}{2}(\mathrm{LL} + \mathrm{LH} + \mathrm{HL} + \mathrm{HH}) = a$, the top-left pixel of each block, which is exactly what a stride-2 1×1 convolution reads.

Placement. In R2 every stride-2 operation of the encoder is replaced by a WaveletPool placed *in front of the original pretrained layer*, whose stride is set to 1:

Replaced operation	Becomes	Initialisation (exact at step 0?)
stem max-pool 3×3, $s = 2$	WP_{64}	average (no)
layer$\{2,3,4\}$.0 grouped 3×3, $s = 2$	$\mathrm{conv}_{3\times3,s=1} \circ \mathrm{WP}$	average (no)
layer$\{2,3,4\}$.0 shortcut 1×1, $s = 2$	$\mathrm{conv}_{1\times1,s=1} \circ \mathrm{WP}$	top-left pixel (**yes**)

All pretrained weights are kept and the cardinality remains 32. At initialisation R2's features have cosine similarity 1.000, 0.909, 0.931, 0.945, 0.955 to R0's at the five stages.

WaveletUnpool. In the decoder,

$$\mathrm{WU}(x) = \mathrm{IDWT}\left([\,\omega_{c,k}\, x_c\,]_{c,k}\right), \qquad \omega_c = (2, 0, 0, 0) \text{ at init}, \tag{6}$$

which at initialisation is nearest-neighbour upsampling. R2's upsampling block is $\mathrm{conv}_{3\times3} \circ \mathrm{WU}$, against R0's $\mathrm{conv}_{3\times3} \circ$ bilinear: the two differ only in the $2\times$ kernel.

Seed pairing. The wavelet parameters are initialised deterministically and draw nothing from the random stream, and the decoder and head are constructed before the encoder. At a fixed seed, every weight shared by two rungs is bitwise identical.

3.5 Decoder and head

A U-Net decoder [6] with explicit skip connections:

$$d_\ell = \phi_\ell([\,\mathrm{Up}_\ell(d_{\ell-1})\,;\,f_{4-\ell}\,]), \qquad d_0 = f_4, \tag{7}$$

where ϕ_ℓ is two 3×3 Conv–BN–ReLU layers and f_{-1} is empty (no skip at full resolution). Widths are 512, 256, 128, 64, 32 at resolutions $32^2, 64^2, 128^2, 256^2, 512^2$. A 1×1 convolution produces one logit per pixel; the sigmoid is applied only in the loss and the metrics.

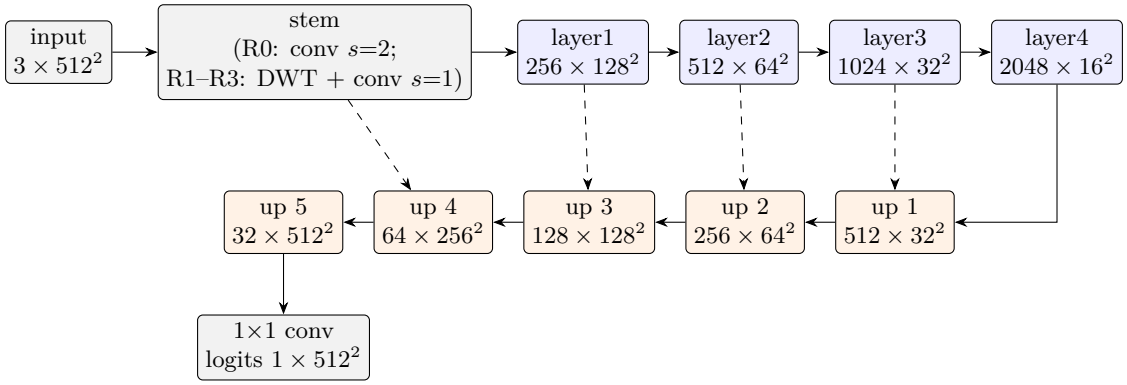


Figure 1: The shared encoder–decoder. Dashed arrows are skip connections. R0 and R1 upsample bilinearly; R2 and R3 replace every stride-2 encoder operation by WaveletPool and every decoder upsampling by WaveletUnpool.

3.6 Parameter counts

Table 3: Trainable parameters (exact). The wavelet rungs add 0.06–0.12% to R0.

Model	Encoder	Decoder + head	Total
R0	22,979,904	23,523,233	46,503,137
R1	23,008,128	23,523,233	46,531,361
R2, R3	23,022,720	23,535,265	46,557,985
U-Net (from scratch)		—	31,037,633
D-LinkNet (ResNet-34)		—	31,095,137

Baselines. U-Net: four levels of widths 64–512 with a 1024-wide bottleneck, trained from scratch. D-LinkNet [7]: ResNet-34 encoder (ImageNet), a centre block of *cascaded* dilated 3×3 convolutions with dilations 1, 2, 4, 8 whose outputs are summed with the input ($d_1 = \rho(c_1(x))$, $d_{k+1} = \rho(c_{k+1}(d_k))$, $\mathrm{out} = x + \sum_k d_k$), and a LinkNet decoder with additive skips.

4 Training

4.1 Loss

Let z be the logits, $p = \sigma(z)$, $y \in \{0, 1\}$ the mask and $m \in \{0, 1\}$ the validity mask, for one 512×512 crop; sums run over pixels.

$$\mathcal{L}_{\text{BCE}} = \frac{\sum m \,\text{bce}(z, y)}{\max(\sum m,\, 1)} \quad \text{(normalised over the valid pixels of the whole batch),} \tag{8}$$

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum p\, y\, m + 1}{\sum p\, m + \sum y\, m + 1} \quad \text{(per sample, averaged over the batch).} \tag{9}$$

R0, R1, R2 and both baselines minimise $\mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}}$. The Dice smoothing constant 1 keeps road-free crops (the majority) numerically stable.

clDice (R3). With a differentiable soft skeleton $S(\cdot)$ (iterated soft erosion/dilation by 3×3 min/max pooling, 6 iterations [8]), on masked inputs $\tilde{p} = p\, m$, $\tilde{y} = y\, m$:

$$T_{\text{prec}} = \frac{\sum S(\tilde{p})\, \tilde{y}}{\sum S(\tilde{p}) + \epsilon}, \quad T_{\text{sens}} = \frac{\sum S(\tilde{y})\, \tilde{p}}{\sum S(\tilde{y}) + \epsilon}, \quad \text{clDice} = \frac{2\, T_{\text{prec}} T_{\text{sens}}}{T_{\text{prec}} + T_{\text{sens}} + \epsilon}, \tag{10}$$

with $\epsilon = 10^{-6}$, and

$$\mathcal{L}_{\text{R3}} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}} + \lambda\,(1 - \text{clDice}), \qquad \lambda = 0.5. \tag{11}$$

On a road-free crop $S(\tilde{y}) = 0$, so the term equals 1 with zero gradient. All losses are computed in fp32; the forward pass runs under bf16 autocast.

4.2 Data pipeline

- **Crops.** Random 512×512 crops, $8\times$ the number of training images per epoch (8864 for Massachusetts, 34,864 for DeepGlobe). With probability 0.75 a crop is resampled (up to 12 tries) until it contains road on valid pixels; crops with less than 50% valid pixels are rejected.
- **Augmentation.** Horizontal/vertical flips and 90° rotations; with probability 0.5 an affine warp (rotation $\pm 15°$, scale 0.9–1.1, bilinear for the image, nearest for the masks, reflective borders); with probability 0.5 brightness and contrast jitter in $[0.9, 1.1]$. ImageNet mean/std normalisation is applied once, after augmentation.
- **Pairing.** All randomness comes from one seeded generator per item, and the data loader has its own generator: for a given seed, every rung sees the identical crop stream.

4.3 Optimisation

AdamW, learning rate $\eta_0 = 3 \times 10^{-4}$, weight decay 10^{-4} on all parameters, batch 16, gradient-norm clipping at 1.0 (a non-finite loss or gradient stops the run). Per-iteration schedule with T_w warmup steps (2 epochs) out of T total:

$$\eta(t) = \begin{cases} \eta_0\big(0.01 + 0.99\,\frac{t}{T_w}\big), & t < T_w, \\ \eta_{\min} + \frac{1}{2}(\eta_0 - \eta_{\min})\Big(1 + \cos\frac{\pi(t - T_w)}{T - T_w}\Big), & t \geq T_w, \end{cases} \qquad \eta_{\min} = 0.01\,\eta_0. \tag{12}$$

Massachusetts: 100 epochs of 554 steps ($T = 55{,}400$, $T_w = 1108$). DeepGlobe: 40 epochs of 2179 steps ($T = 87{,}160$, $T_w = 4358$). One recipe for every model; nothing was tuned per rung.

4.4 Model selection

After each epoch, the pooled validation IoU is computed at threshold 0.5 on non-overlapping tiles; the best epoch's weights are kept. After training, the decision threshold is chosen on the validation split from $\{0.20, 0.25, \ldots, 0.80\}$ to maximise pooled IoU. The test split is not read during training or selection.

5 Evaluation

5.1 Full-image inference

Each test image is covered by 512×512 tiles at stride 256, with the last row and column aligned to the image edge (25 tiles per Massachusetts image, 9 per DeepGlobe image). Tile probabilities are averaged where tiles overlap,

$$P(u) = \frac{\sum_t \mathbf{1}[u \in t]\, P_t(u)}{\sum_t \mathbf{1}[u \in t]}, \tag{13}$$

and the prediction is $\hat{Y} = \{u : P(u) \geq \tau^*\}$. All metrics are computed on the stitched full-resolution image, over valid pixels only. Each run's test split is scored exactly once; the probability maps are saved.

5.2 Metrics

With $\mathrm{TP}, \mathrm{FP}, \mathrm{FN}$ counted over valid pixels:

- **IoU** $= \mathrm{TP}/(\mathrm{TP}+\mathrm{FP}+\mathrm{FN})$, reported per image (mean over images) and pooled (counts summed over images). An image with no valid pixels has no score; an all-background prediction on a road-free image scores 1.
- **Relaxed precision/recall/F1** [2]: a predicted pixel is correct if a ground-truth road pixel lies within Euclidean distance $\rho = 3$ px, and vice versa.
- **clDice** (hard skeletons) [8].
- **APLS** [9]: graphs are built from the skeletons of prediction and ground truth; for 500 sampled node pairs per direction, $\mathrm{APLS} = 1 - \frac{1}{N}\sum \min\!\left(1, \frac{|L-\hat{L}|}{L}\right)$, averaged over both directions; a node with no counterpart within 8 px contributes the maximum penalty.

6 Statistical analysis

Let $u_{a,s,i}$ be the test IoU of image i for arm $a \in \{0, 1\}$ (baseline, comparison) and seed s; $\bar{u}_{a,i}$ its mean over the arm's seeds; N the number of images. The point estimate is

$$\hat{\Delta} = \frac{1}{N}\sum_{i=1}^{N} (\bar{u}_{1,i} - \bar{u}_{0,i}). \tag{14}$$

Resampling seeds as well as images targets the training procedure rather than three particular trained models. With three seeds the interval is crude: in simulation it covered 0.94–0.98 in most regimes but 0.86 when seed variance dominated at $N = 935$.

Algorithm 1 Two-level paired bootstrap (primary interval)

for $b = 1, \ldots, B$ ($B = 10{,}000$, fixed seed) **do**
  draw seeds S_0^*, S_1^* with replacement, independently within each arm
  draw image indices I^* with replacement (the same draw for both arms)
  $\Delta_b^* \leftarrow \frac{1}{N} \sum_{i \in I^*} \left(\frac{1}{|S_1^*|} \sum_{s \in S_1^*} u_{1,s,i} - \frac{1}{|S_0^*|} \sum_{s \in S_0^*} u_{0,s,i} \right)$
end for
95% CI: the 2.5th and 97.5th percentiles of $\{\Delta_b^*\}$
$p = \min\left(1,\ 2\min\left(\frac{\#\{\Delta_b^* \le 0\}+1}{B+1},\ \frac{\#\{\Delta_b^* \ge 0\}+1}{B+1}\right)\right)$

Seed-paired t interval. Seeds are paired by design (same crop stream and same shared initialisation). With $d_s = \frac{1}{N} \sum_i (u_{1,s,i} - u_{0,s,i})$ over the $n = 3$ common seeds, the interval is $\bar{d} \pm t_{0.975,\, n-1}\, s_d / \sqrt{n}$.

Decision rule (pre-specified). A difference is claimed only if *both* intervals exclude zero with the same sign. If exactly one excludes zero, or they exclude it with opposite signs, the result is *inconclusive* at three seeds (and more seeds would be needed before any claim). If neither excludes zero: *no detected difference*.

7 Results

All 36 runs completed (no failures). Test scores are reported at the validation-selected threshold.

7.1 Primary comparison

Table 4: Primary analysis: R1 (DWT at the input) versus R0 (stock), per-image test IoU, 3 seeds per arm.

Dataset	N	R0 pooled IoU	R1 pooled IoU	$\hat{\Delta}$	Two-level 95% CI (p)	Seed-paired t 95% CI
Massachusetts	49	0.6562	0.6554	+0.0004	$[-0.0040, +0.0060]$ (0.90)	$[-0.0095, +0.0104]$
DeepGlobe	935	0.6771	0.6785	+0.0017	$[-0.0018, +0.0059]$ (0.38)	$[-0.0072, +0.0106]$

Decision: no detected difference on either dataset. The robustness checks agree: the difference in pooled IoU is -0.0008 $[-0.0044, +0.0032]$ (Massachusetts) and $+0.0015$ $[-0.0013, +0.0045]$ (DeepGlobe); in IoU at a fixed threshold of 0.5 it is $+0.0014$ $[-0.0032, +0.0077]$ and $+0.0018$ $[-0.0017, +0.0061]$. No test image has fewer than 2000 road pixels on valid pixels, so the near-empty-image sensitivity analysis excludes nothing.

7.2 Secondary comparisons

7.3 Other metrics (exploratory)

7.4 Compute

All 36 runs trained on single NVIDIA L40S GPUs (48 GB) with bf16 autocast. At batch 16 of 512^2 crops, the GPU self-test measured 3.65–3.89 steps/s for R0–R3 (peak memory 11.9–13.0 GB), 2.97 steps/s for U-Net (17.2 GB) and 9.69 steps/s for D-LinkNet (5.4 GB). A Massachusetts epoch of R0 took about 132 s including validation (≈3.7 h per run); a DeepGlobe epoch took 9.1–9.9 min (≈6.1–6.6 h per run).

Table 5: All models against R0 (per-image test IoU). Secondary and exploratory: no multiplicity correction.

Dataset	Model	Pooled IoU (mean ± sd)	$\hat{\Delta}$ vs R0	Two-level 95% CI	Decision
Massachusetts	R0	0.6562 ± 0.0011	—	—	—
	R1	0.6554 ± 0.0024	$+0.0004$	$[-0.0040, +0.0060]$	no difference
	R2	0.6545 ± 0.0001	-0.0009	$[-0.0060, +0.0049]$	no difference
	R3	0.6553 ± 0.0018	-0.0018	$[-0.0061, +0.0020]$	no difference
	U-Net	0.6632 ± 0.0001	$+0.0095$	$[+0.0047, +0.0151]$	difference
	D-LinkNet	0.6515 ± 0.0014	-0.0043	$[-0.0114, +0.0023]$	inconclusive
DeepGlobe	R0	0.6771 ± 0.0023	—	—	—
	R1	0.6785 ± 0.0010	$+0.0017$	$[-0.0018, +0.0059]$	no difference
	R2	0.6823 ± 0.0011	$+0.0039$	$[+0.0001, +0.0084]$	inconclusive[†]
	R3	0.6743 ± 0.0013	-0.0027	$[-0.0069, +0.0019]$	no difference
	U-Net	0.6461 ± 0.0008	-0.0332	$[-0.0381, -0.0277]$	difference
	D-LinkNet	0.6655 ± 0.0005	-0.0081	$[-0.0126, -0.0033]$	inconclusive

[†] The bootstrap interval excludes zero but the seed-paired t interval $[-0.0059, +0.0137]$ does not (per-seed differences $+0.0033, +0.0003, +0.0082$).

Table 6: Mean ± sd over 3 seeds of per-image test metrics. Exploratory; the metrics are correlated.

Dataset	Model	IoU	Relaxed F1 (ρ=3)	clDice	APLS
Massachusetts	R0	0.6529 ± 0.0008	0.9072 ± 0.0011	0.8812 ± 0.0012	0.8047 ± 0.0041
	R1	0.6533 ± 0.0032	0.9087 ± 0.0037	0.8821 ± 0.0036	0.8029 ± 0.0040
	R2	0.6520 ± 0.0034	0.9079 ± 0.0048	0.8822 ± 0.0047	0.8029 ± 0.0039
	R3	0.6511 ± 0.0016	0.9060 ± 0.0012	0.8828 ± 0.0016	0.8040 ± 0.0027
	U-Net	0.6624 ± 0.0003	0.9155 ± 0.0002	0.8895 ± 0.0004	0.8006 ± 0.0030
	D-LinkNet	0.6486 ± 0.0014	0.9080 ± 0.0006	0.8806 ± 0.0008	0.8002 ± 0.0025
DeepGlobe	R0	0.6751 ± 0.0032	0.8978 ± 0.0026	0.8665 ± 0.0020	0.7755 ± 0.0045
	R1	0.6768 ± 0.0005	0.8988 ± 0.0004	0.8670 ± 0.0007	0.7777 ± 0.0020
	R2	0.6790 ± 0.0011	0.8996 ± 0.0005	0.8687 ± 0.0003	0.7814 ± 0.0013
	R3	0.6723 ± 0.0010	0.8952 ± 0.0013	0.8683 ± 0.0009	0.7763 ± 0.0045
	U-Net	0.6419 ± 0.0009	0.8764 ± 0.0008	0.8378 ± 0.0011	0.7193 ± 0.0020
	D-LinkNet	0.6669 ± 0.0008	0.8955 ± 0.0004	0.8613 ± 0.0009	0.7614 ± 0.0036

8 Discussion

The DWT front-end does not help a pretrained ResNeXt. On both datasets the primary interval is narrow and centred near zero; gains larger than about 0.6 IoU points are excluded. Proposition 1 explains why this is unsurprising: a DWT followed by a convolution is a reparameterised strided convolution with a larger footprint, so it adds no information that a pretrained first layer lacks. Starting R1 exactly at R0 isolates this effect; with the non-equivalent "LL-only" adaptation, any difference would have been confounded with the loss of pretrained features.

Learned wavelet pooling throughout (R2). On DeepGlobe, R2 is ahead of R0 on every metric (IoU +0.004, APLS +0.006) with smaller seed-to-seed spread, but the pre-specified rule classifies it as inconclusive at three seeds, and it shows nothing on Massachusetts. R2 replaces aliasing strided operations by learned filters that start as averaging, i.e. anti-aliased downsampling [10, 11]; if it has a benefit, robustness to shifts and corruptions is the more likely place to find it than clean IoU. Any

such test should be specified before it is run.

Topology loss (R3). Adding soft-clDice did not improve IoU, clDice or APLS on either dataset.

Baselines. The from-scratch U-Net is the best model on Massachusetts but the worst on DeepGlobe, suggesting that ImageNet pretraining matters less on Massachusetts' smaller training set and near-constant-width centre-line labels than on DeepGlobe's more varied scenes.

9 Limitations

- Three seeds per model; the headline interval can be anti-conservative when seed variance dominates. Only R1 − R0 per-image IoU was pre-specified.
- R2 and R3 change the encoder and the decoder operators together.
- Only the exact stem initialisation was run; the "LL-only" variant was not.
- One untuned recipe for every model; weight decay is applied to all parameters including BatchNorm.
- Checkpoint and threshold are selected per run on validation, which is 14 images for Massachusetts.
- Massachusetts scores exclude a custom no-data mask and are not comparable with published numbers. The DeepGlobe split is our own random split by image (adjacent tiles may fall in different splits), and its files come from a mirror whose pixel identity with the original release was not verified.
- bf16 autocast and non-deterministic GPU kernels: reruns are statistically, not bitwise, reproducible (all runs used the same GPU model, the NVIDIA L40S).
- Edge-aligned tiling with uniform overlap averaging; no test-time augmentation.
- The soft skeleton uses a 3×3 square min-pool rather than a cross.

10 Reproducibility

The experiment ran from a frozen snapshot of the code (commit `c1d1e2f`) on 12–14 October 2026. Each run directory records the resolved configuration, seed, code commit, library versions, CUDA/cuDNN versions, the SHA-256 of the pretrained weights, the DeepGlobe split fingerprint, the training history, the validation-selected threshold, and per-image test scores. The code base includes a 458-test suite, a CPU end-to-end rehearsal and a GPU self-test; the decisions taken during implementation, with their reasons, are recorded in `docs/DECISIONS.md`.

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